

Causal Survival Inference from Reconstructed Data: Design-Based and Model-Based Bayesian Nonparametric Approaches

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Hopkins

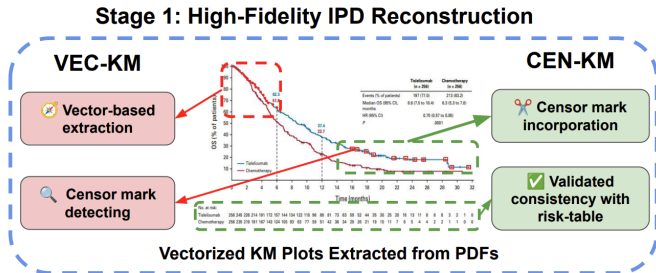
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Roadmap

- ① Motivation and problem setup
- ② Methodology
- ③ Simulation study
- ④ Real data application
- ⑤ Results and conclusions

Motivation

- Many clinical studies provide only Kaplan-Meier curves, not raw individual patient data (IPD).
- Lang et al. developed **RESOLVE-IPD** to reconstruct high-fidelity IPD from KM curves.



- VEC-KM**: extracts vectorized KM curves.
- CEN-KM**: reconstructs IPD.

- But reconstruction is only the first step.
- Our question is: after reconstructing IPD, can we perform **causally interpretable** survival inference?
- This is challenging because reconstructed data still suffer from:
 - **Missing covariates:** Kaplan–Meier reconstruction provides (T_i, δ_i, A_i) , but not individual-level covariates X_i , which are important for causal inference.
 - **Unclear target population:** different trials involve different patient populations, so we cannot simply pool them together. A naive pooled estimate may not have a clear causal interpretation.
- **Goal:** move from *data reconstruction* to *causal survival inference*

Problem Setup

- For each reconstructed individual:
 - T_i = observed survival time
 - $\delta_i \in \{0, 1\}$ = event indicator (1 = event, 0 = censored)
 - $A_i \in \{0, 1\}$ = treatment assignment
- Survival probability under treatment a at time t_0 :

$$S_a(t_0) = \Pr(T^{(a)} > t_0)$$

- Causal estimand:

$$\Delta(t_0) = S_1(t_0) - S_0(t_0)$$

- **Goal:** estimate $\Delta(t_0)$.

Step 1: Covariate Augmentation

- We augment each reconstructed individual with pseudo-covariates X_i using published trial-level summaries.

- Examples:

- **Sex:** if the reported male proportion is p , generate

$$X_i^{(\text{male})} \sim \text{Bernoulli}(p).$$

- **Age:** if the reported mean age is μ , generate

$$X_i^{(\text{age})} \sim N(\mu, \sigma^2),$$

- This gives an augmented dataset

$$(T_i, \delta_i, A_i, X_i),$$

- Limitation: the marginals are matched, but the joint dependence structure is not recovered.

Step 2: Reweight Target Population

- Suppose we combine two reconstructed trials:
 - Trial 1: younger patients, mean age 55, estimated survival contrast $\Delta(t_0) = 0.30$
 - Trial 2: older patients, mean age 70, estimated survival contrast $\Delta(t_0) = 0.10$
- A naive pooled estimated survival contrast is $\Delta_{\text{pooled}}(t_0) \approx 0.20$.
- But what does 0.20 mean?
 - It is not the effect for younger patients
 - It is not the effect for older patients
 - It only reflects the **mixture of trial populations** in the pooled data
- If our target population is patients with a mean age 65, we should **reweight** the individuals to match this target population.

Step 2: Reweight Target Population

- Let F_X^* denote the target population.
- Choose weights $\{w_i\}_{i=1}^n$ to match the target:

$$\sum_{i=1}^n w_i g(X_i) \approx m^*, \quad \sum_{i=1}^n w_i = 1.$$

- The treatment survival probability at time t_0 is

$$S_a^*(t_0) = \sum_{i:A_i=a} w_i \mathbf{1}(T_i > t_0),$$

- The causal contrast is

$$\Delta^*(t_0) = S_1^*(t_0) - S_0^*(t_0).$$

- The effective sample size (ESS) to assess weighting stability.

$$\text{ESS} = \frac{1}{\sum_{i=1}^n w_i^2}$$

Step 2: Why Is Reweighting Not Enough?

- Suppose after reweighting to the target population, we estimate

$$\Delta^*(t_0) = 0.20.$$

- But what does this number really mean?
 - Is 0.20 stable or highly variable?
 - Would it change under another reasonable weighting scheme?
 - Would it change if we model the survival process more explicitly?
- Reweighting gives a **point estimate**.
- We need to quantify uncertainty.

Step 3: Why Bayesian Nonparametrics?

- We introduce **Bayesian nonparametric (BNP)** methods to get a posterior distribution for the treatment effect $\Delta^*(t_0)$.
- **where does the uncertainty come from?**
- We propose two BNP methods:
 - **Design-based BNP:** uncertainty comes from weighting scheme.
 - **Model-based BNP:** uncertainty comes from the survival model.

Step 3: Design-Based Bayesian Nonparametrics

- We treat the sample weights as random:

$$w = (w_1, \dots, w_n) \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_n).$$

- The sampled weights must satisfy:
 - $w_i \geq 0$ and $\sum_{i=1}^n w_i = 1$.
 - $\sum_{i=1}^n w_i g(X_i) = m^*$, to match the target population
 - reasonable ESS.
- For each posterior draw $w^{(s)}$, compute

$$S_a^{(s)}(t_0) = \sum_{i:A_i=a} w_i^{(s)} \mathbf{1}(T_i > t_0),$$

and

$$\Delta^{(s)}(t_0) = S_1^{(s)}(t_0) - S_0^{(s)}(t_0).$$

- The collection of $\Delta^{(s)}(t_0)$ gives the posterior mean and credible interval.

Step 3: Model-Based Bayesian Nonparametrics

- We model the **survival process** itself.
- Time is partitioned into intervals, and each interval has a hazard increment:

$$\lambda_{a,j} \sim \text{Gamma}(\alpha, \beta).$$

$$\lambda_{a,j} \mid \text{data} \sim \text{Gamma}(\alpha + D_{a,j}, \beta + Y_{a,j})$$

- Here, $D_{a,j}$ is the number of events in arm a during interval j , and $Y_{a,j}$ is the total exposure time in that interval.
- These hazard increments determine the survival curve:

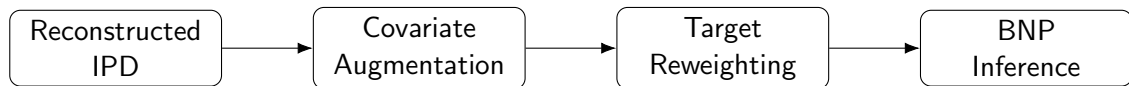
$$S_a(t) = \exp \left(- \sum_{j:t_j \leq t} \lambda_{a,j} \Delta t_j \right).$$

- Posterior sampling of $\lambda_{a,j}$ gives posterior draws of

$$\Delta^{(s)}(t_0) = S_1^{(s)}(t_0) - S_0^{(s)}(t_0).$$

Methodology Overview

- Input: reconstructed individual patient data (IPD)
- Step 1: augment missing covariates using trial-level summaries
- Step 2: reweight pooled data to a target population
- Step 3: estimate treatment effect using Bayesian nonparametric methods



Simulation Study Overview

- Goal: evaluate the full pipeline under controlled settings
- Generate synthetic multi-trial survival data:
 - heterogeneous covariate distributions across trials
 - treatment randomized within each trial
- Only Kaplan–Meier curves and summary statistics are observed
- Apply full pipeline:
 - reconstruct IPD
 - augment covariates
 - reweight to target population
 - estimate $\Delta(t_0)$
- Evaluate:
 - reconstruction accuracy
 - bias and variability of estimates
 - stability under covariate shift

Simulation Results: Validation and Heterogeneity

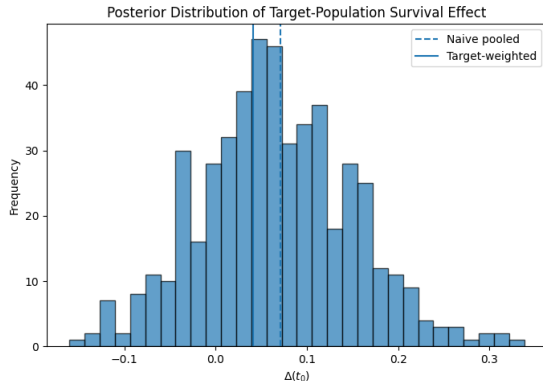
- **Single-trial validation:**

- reconstructed IPD closely matches true survival curves
- negligible error in survival estimates at t_0

- **Multi-trial setting:**

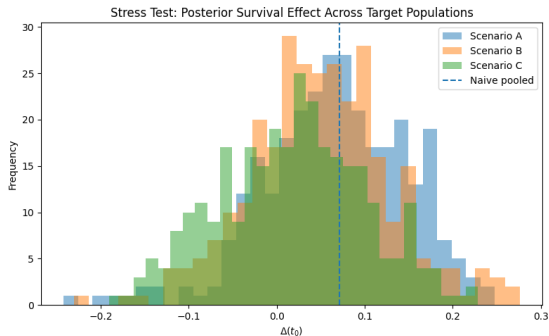
- trials have different covariate distributions
- treatment effects vary across trials

- Naive pooling averages across heterogeneous populations



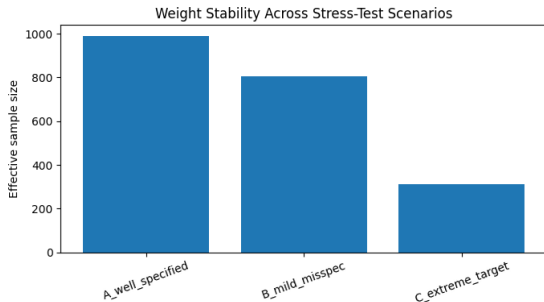
Simulation Results: Stress Testing Target Populations

- Evaluate robustness under increasing covariate shift
- **Findings:**
 - posterior uncertainty increases with shift
 - estimates become more variable
 - extreme targets are less stable
- Reweighting is most reliable near observed data support



Stress Testing: Weight Stability

- Effective sample size (ESS) diagnoses weighting stability
- As the target population moves farther from observed data:
 - weights become more concentrated
 - ESS decreases
 - inference becomes less reliable



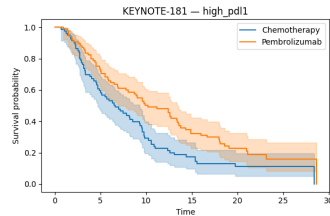
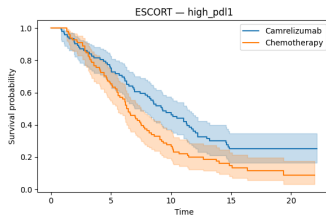
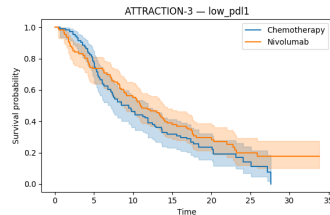
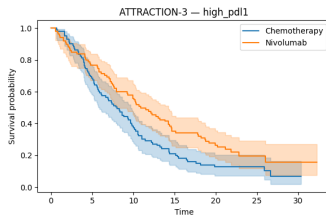
Real Data Application: KM-GPT and RESOLVE-IPD

- Reconstructed oncology trial datasets:

- ATTRACTION-3
- ESCORT
- KEYNOTE-181

- Compare downstream causal inference under:

- KM-GPT
- RESOLVE-IPD



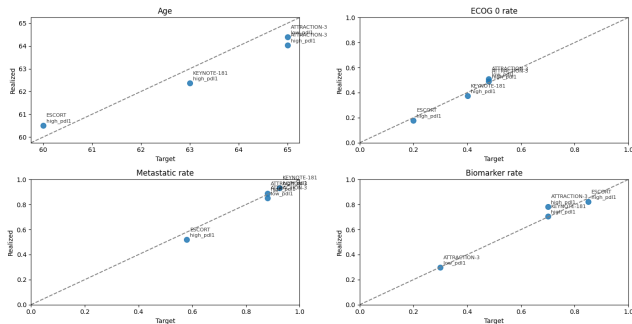
Real Data Application: KM-GPT and RESOLVE-IPD

- Two reconstruction pipelines used as inputs:
- **KM-GPT:**
 - automated extraction and reconstruction from KM curves
 - produces individual-level survival times and events
- **RESOLVE-IPD:**
 - reconstruction using risk tables and censoring information
 - incorporates additional structural constraints
- **Key question:**
 - how sensitive are downstream causal estimates to reconstruction method?

Covariate Augmentation Validation

- Augmented covariates are generated from trial-level summaries
- Validation:
 - compare target summaries
 - compare realized sample moments
- Strong agreement supports use in downstream weighting

Covariate Augmentation Checks: Target vs Realized Summaries



Main Results: Reconstruction Pipelines and Weighting

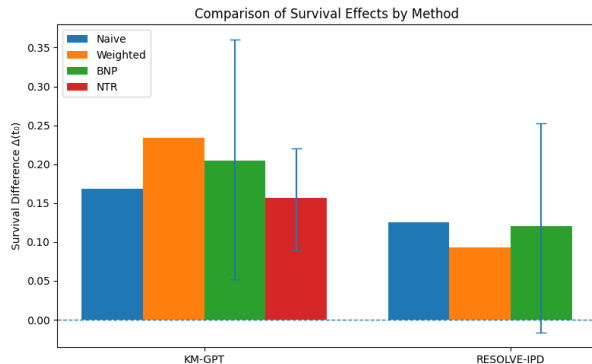
- **Naive estimates:**

- differ across KM-GPT and RESOLVE-IPD
- sensitive to reconstruction method

- **Target-population weighting:**

- substantially changes $\Delta(t_0)$
- can increase or decrease effect size

- Reconstruction + weighting both materially affect inference



Main Results: BNP Uncertainty and NTR Hazard Model

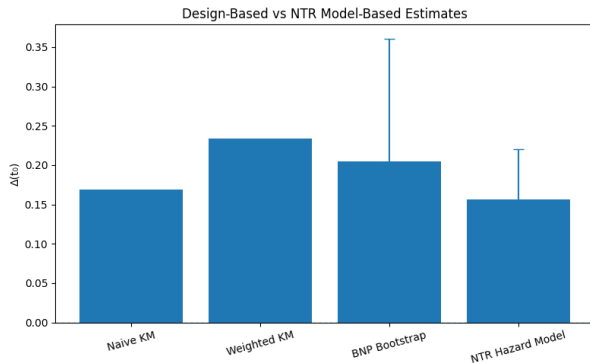
- **Design-based BNP:**

- posterior uncertainty differs by reconstruction pipeline

- **Model-based BNP (NTR hazard model):**

- produces moderated (smoothed) estimates
- models cumulative hazard directly
- reduces sensitivity to reconstruction noise

- Modeling assumptions materially affect conclusions



Key Takeaways

- Reconstruction method affects downstream causal estimates
- Target-population weighting is essential for defining interpretable estimands
- Bayesian nonparametric methods provide meaningful uncertainty quantification
- Design-based and hazard-based model approaches can yield systematically different estimates due to smoothing and modeling assumptions

- **Limitations:**

- covariates are simulated from aggregate summaries
- joint dependence between covariates is not fully captured
- results depend on reconstruction quality of IPD
- hazard-based modeling introduces smoothing that may attenuate local features of the survival curve

- **Future work:**

- incorporate richer covariate modeling (e.g., joint distributions)
- extend to time-varying treatments and covariates
- explore alternative Bayesian nonparametric survival models
- apply to broader clinical and observational datasets